

Air Series
User Guide

2026/07/13

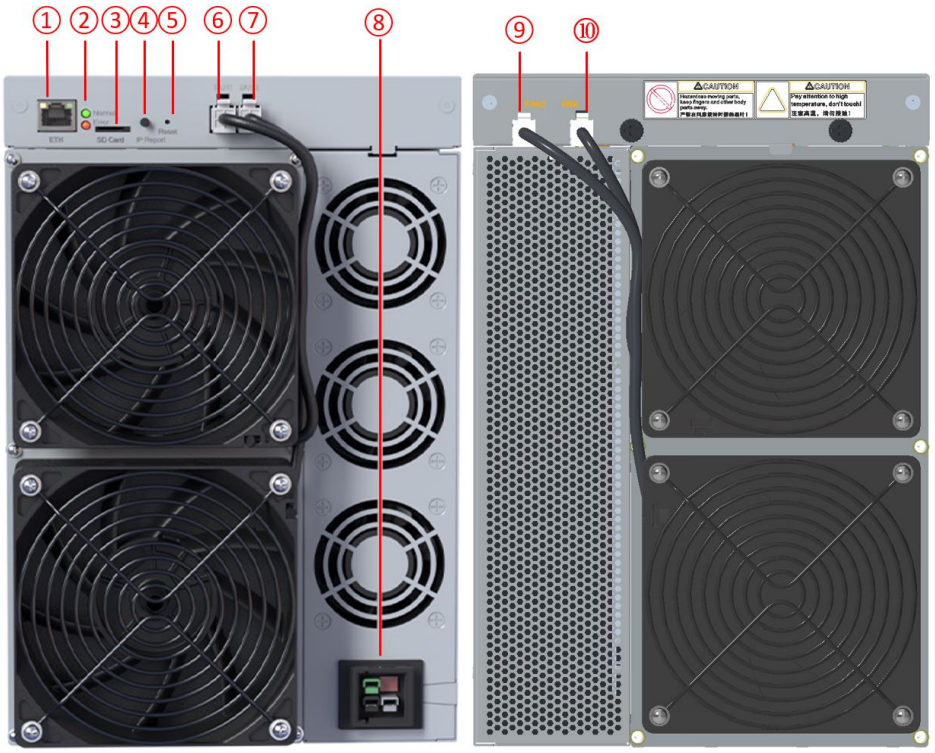
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1. Product Introduction

Applicable models for this manual: A2 Air, A2 Pro Air, A3 Air, A3 Pro Air, A4 Pro Air (The parameters in this guide are applicable to the general conditions of the listed models. If there are differences in technical parameters for specific models, please refer to the latest data on the official Bitdeer website. If you have any questions or need further assistance, please contact the Bitdeer customer support team.)



1.1 Component Reference

This manual applies to the SealMiner AX Air series air-cooled servers.

① Ethernet port
② Indicator light
③ SD card slot
④ Find IP button
⑤ Reset button
⑥ FAN1 port
⑦ FAN2 port
⑧ Power port

⑨ FAN3 port
⑩ FAN4 port

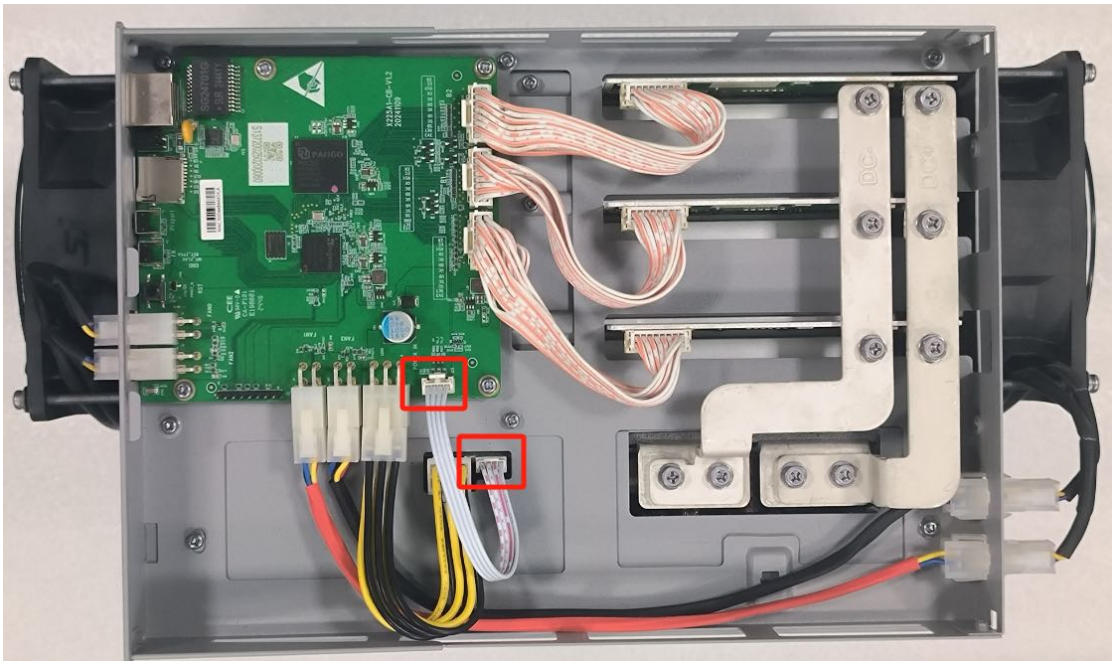
2. Server wiring and shelf operation safety reminder

2.1 Server Wiring

Connect the control board cables to the fan, power supply, and computing board correctly according to the instructions. Make sure the connector clips are locked. Do not force the cable plugs in backwards. Inserting them incorrectly can damage the control board and signal cables when you turn on the power.

2.2 Power control cable connection

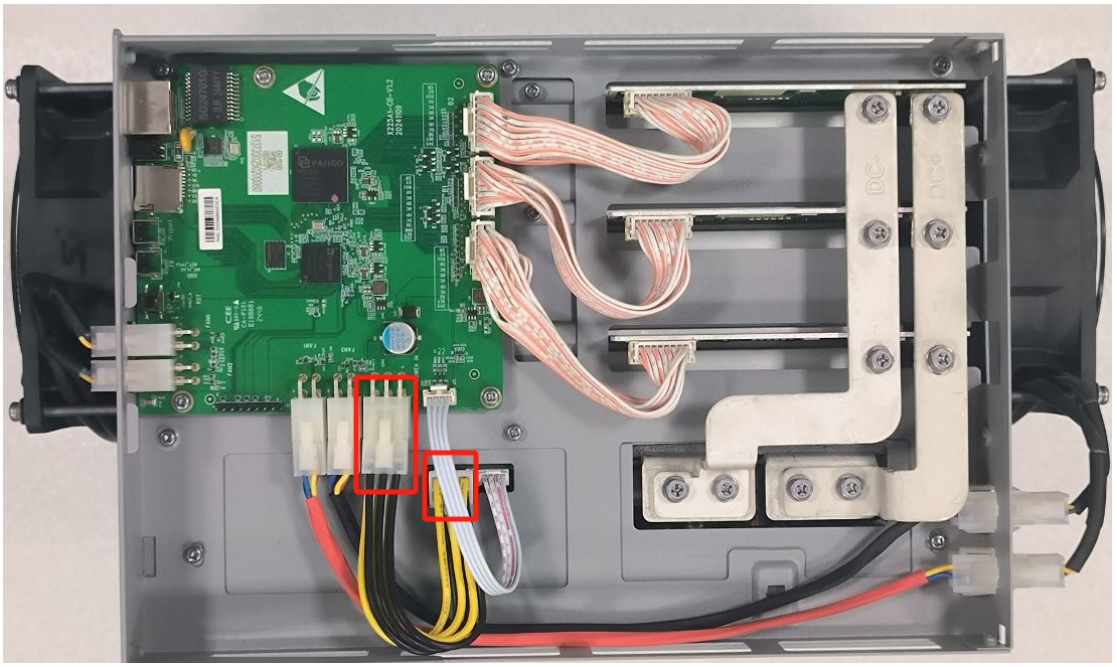
The 4-pin power IIC control cable features a locking buckle on the plug. Ensure correct orientation during connection as shown in the figure below, and verify that the buckle is securely engaged after insertion.



2.3 Precautions for connecting the power cable of the control board

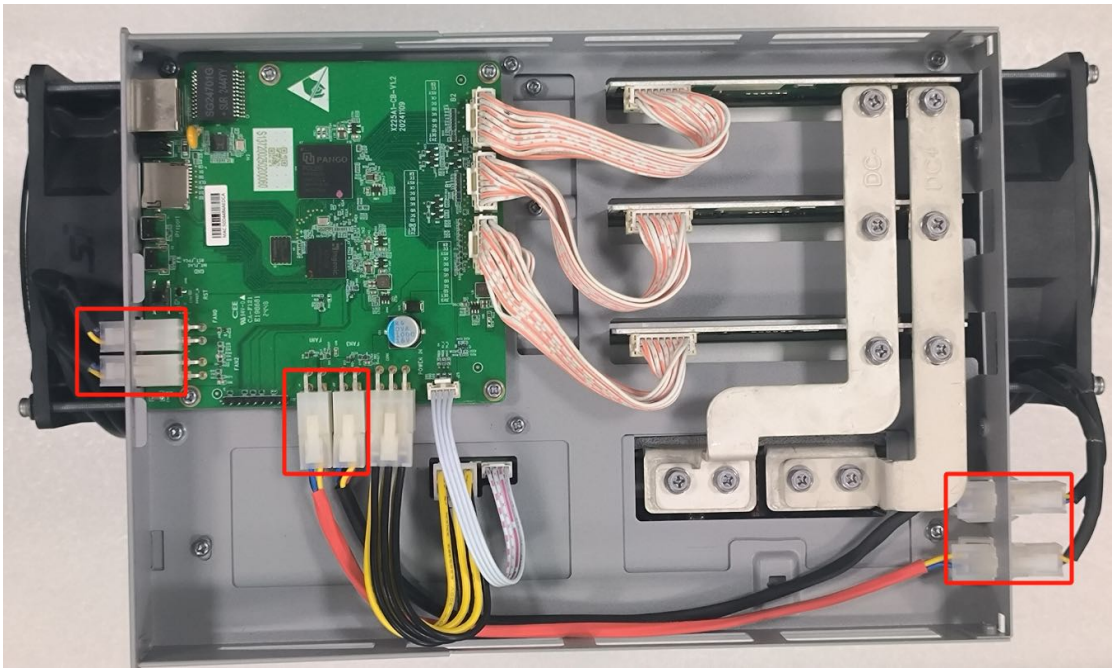
The control board power cable is a 2x3PIN connector with a locking buckle. Please refer to the figure below for the correct connection

method. Ensure the insertion direction is accurate and the buckle is securely latched after connection.



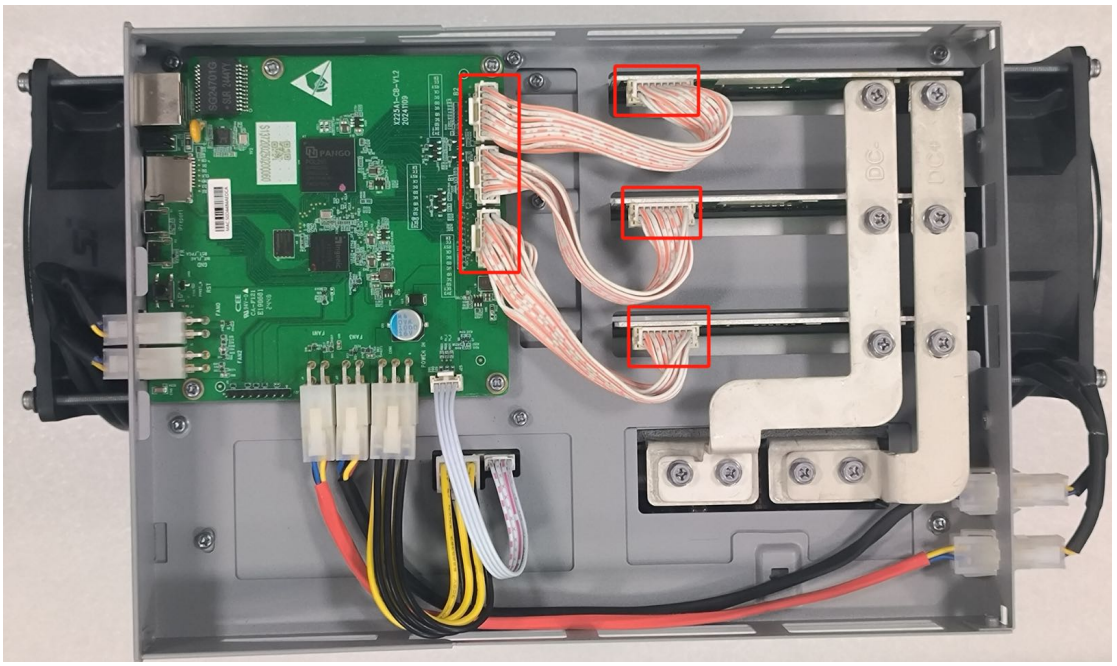
2.4 Fan cable connection

The fan cable is a 2*2PIN with a locking buckle. Please refer to the illustration below for the correct connection method. Ensure the insertion direction is accurate and the buckle is securely latched after connection.



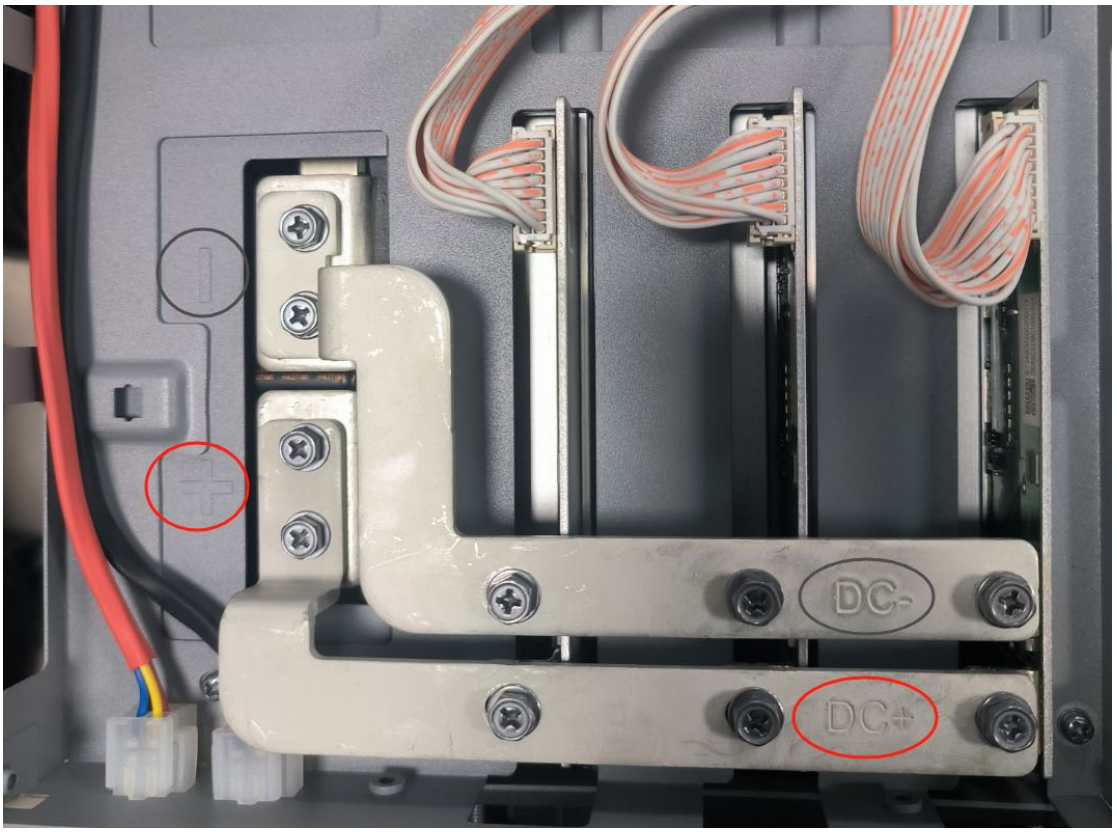
2.5 Precautions for connecting the control board and the hash board

The hashboard cable is a 2*8PIN with a locking buckle. Please refer to the illustration below for the correct connection method. Ensure the insertion direction is accurate and the buckle is securely latched after connection.



2.6 Power Copper Busbar Connection Precautions

During the installation of the hash board copper busbars, please ensure that copper bars of the same polarity are positioned as illustrated below. Verify that the copper bars are placed parallel to each other and are not in contact. Securely tighten the copper bar fixing screws. After tightening, reconfirm that the copper bars remain parallel and without any contact. Failure to adhere to these instructions may result in abnormal operation or even permanent damage to the machine.



2.7 Machine wiring check

After connecting the cables and tightening the screws, ensure that the cables and screws are installed correctly.

2.8 Precautions for server transportation and racking

Please wear work gloves when handling and positioning the server onto the rack. To prevent accidental drops during transportation, minimize manual carrying over long distances. Exercise caution when placing the server on the shelf to avoid hand injuries from metal parts, and ensure cables are not pulled, which could cause the machine to fall.

3. Server configuration and preparation

3.1 Server configuration item list

	Item Name	Qty	Item Use	Remarks
1	Computer	1 unit	Configuring server operations	
2	Network Switch	1 unit	Enabling network communication between the configuration computer and the server	The network switch must be able to connect to the external network.
3	DHCP/Router	1 unit	Providing dynamic IP address for initial server power-on	The server is configured to obtain a dynamic IP address via DHCP by default.

3.2 Server network environment

The factory default setting for the server is to obtain a dynamic IP address through DHCP, hence a DHCP server must be configured in the network, or the router must enable the DHCP dynamic IP address allocation service.

4. Server connection and power-on check

(1) Prior to connecting any cables and powering on the server, thoroughly inspect it to ensure that no heat sinks or other components have become detached. It is imperative that all heat sinks and components are securely in place before proceeding with wiring and power-up.

(2) Connect the power cord to the server and the network cable to the switch. Before powering on the server, verify that the power control cable, fan control cable, control board power supply cable, hash board cables, and fan cables are firmly secured and not loose. Additionally, confirm that the copper busbars are correctly connected.

Note:

1) When installing the hash board copper busbars, ensure the positive and negative polarity busbars are correctly positioned. Verify that the busbars are placed parallel to each other and are not in contact. The copper busbar fixing screws must be securely tightened. After tightening, reconfirm that the copper bars remain parallel and without any contact. Failure to comply with these instructions may lead to abnormal operation or even permanent damage (burnout) of the machine.

2) The power control cable between the control board and the power supply must be connected correctly. Incorrect connection may prevent the control of the power supply voltage output, resulting in a lack of hashing power.

3) The control board must have a reliable connection to the fan cables. If a fan cable becomes detached or poorly connected, the server may be unable to cool down, triggering a protective power-off.

4) Server Data Configuration (Webpage Configuration) (Software)

5) Querying the Dynamic IP Address Obtained by the Server

6) Power Connection: Use the provided 1.2-meter power cord to connect the server to the PDU, ensuring the connector is fully seated in the server's power port (Component 8). The upstream

PDU circuit breaker acts as the disconnection device and must have a rated breaking current of 20A.

5. Server data configuration

5.1 Query the Dynamic IP Address Assigned to the Server

Ensure your computer is connected to the same network segment as the server. Utilize the Sealminer management tool and click "Scan、Detect" to discover and open the server's IP address.(Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)

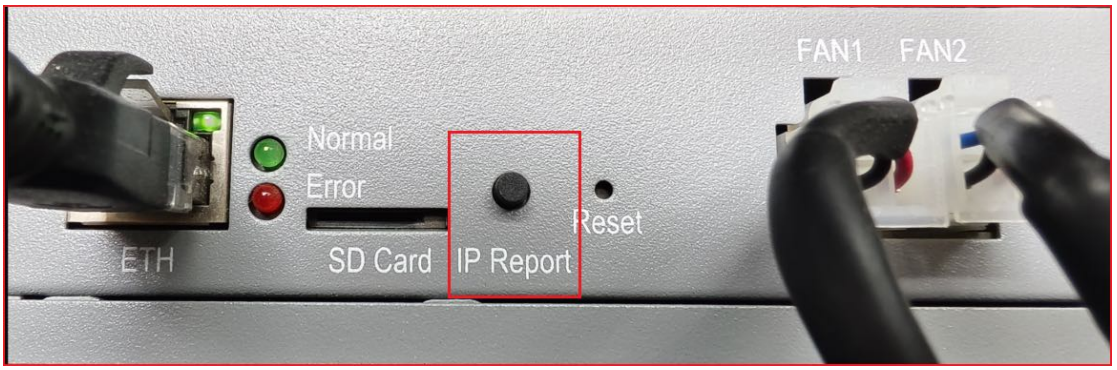




Scan/Detect Window.

Click "Start" on the window to initiate the scan.

IP Reporting Method: After starting the miner, press and hold the IP Report button on the miner for 5 seconds. Wait a few seconds, and the current device's IP address will be displayed in the software. (Note: The PC running the management tool and the miner must be on the same local area network).



View the dynamically obtained IP and MAC addresses reported by the server in the Sealminer management tool.

Note:

1) If none of the indicator lights on the server control board panel are on after powering on, please check the reliability and correctness of the power cable connections.

2) If the indicator light on the right side of the server control board panel is on, but the network port lights are off or the green light is not flashing, please check if the network switch is functioning correctly, if the network cable connection is reliable, and if there are any issues with the network cable quality.

3) The computer running the Sealminer management tool and the server must be on the same network segment. Otherwise, the software may not receive the broadcast packets sent by the server, and consequently, the IP address and MAC address information reported by pressing the server's IP Report button will not be found.

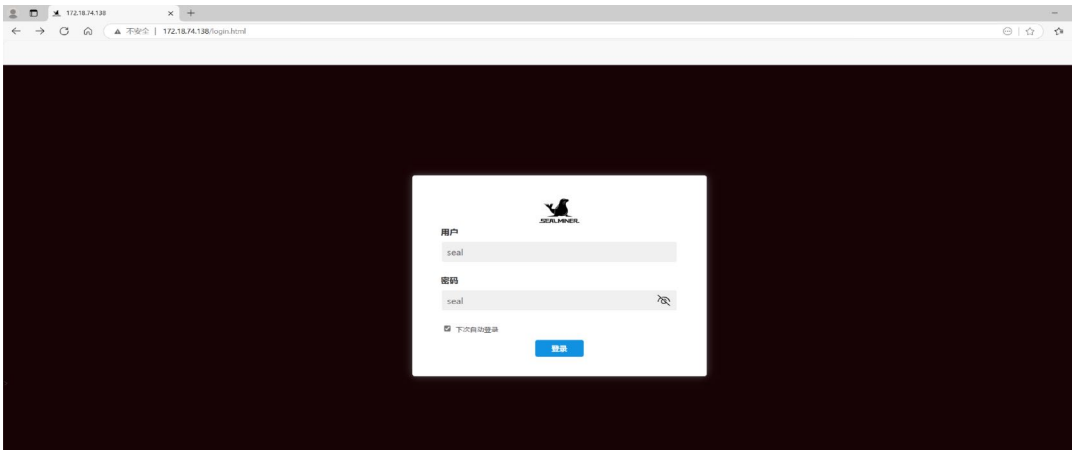
4) If the computer and server are on the same network segment and DHCP service is enabled on the network, but the Sealminer management tool does not find the server's IP after pressing the IP Report button, press and hold the Reset button on the server panel for more than 5 seconds to restore the factory default configuration. Then, power off and then power on the server to restart it. After the server starts normally, press the IP Report button for 5 seconds to detect the server's IP address.

5.2 Mining pool & miner data

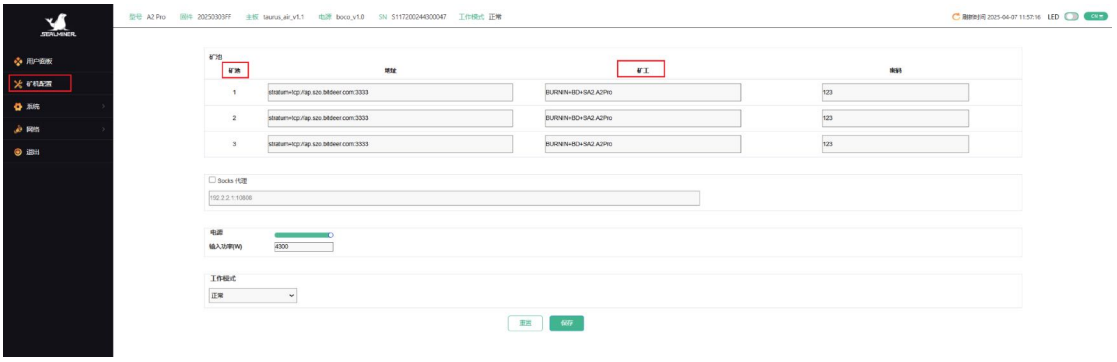
5.2.1 Mining pool & miner configuration

(1) Log in to the web page

Account: seal, Password: seal



(2) In the mining machine configuration interface, modify the mining pool address and miner name. After modification, click "Save" below to save the changes made.(Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)

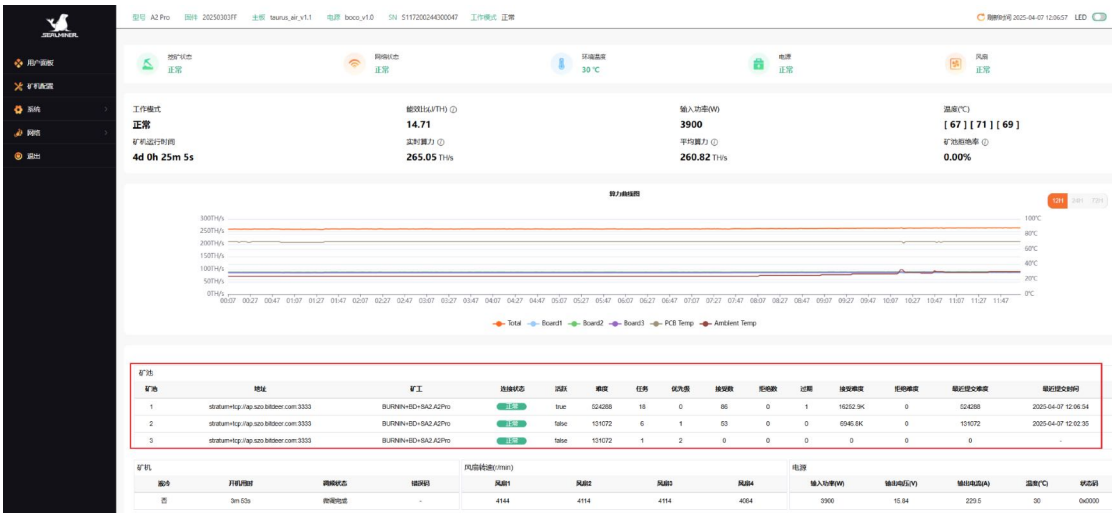


After the mining pool configuration is modified, it will take effect immediately without needing to restart the Miner program or the mining machine.

(3) Check whether the configuration changes are effective.

In the user panel interface, check whether the configured mining pool and miner data can be seen. (Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and

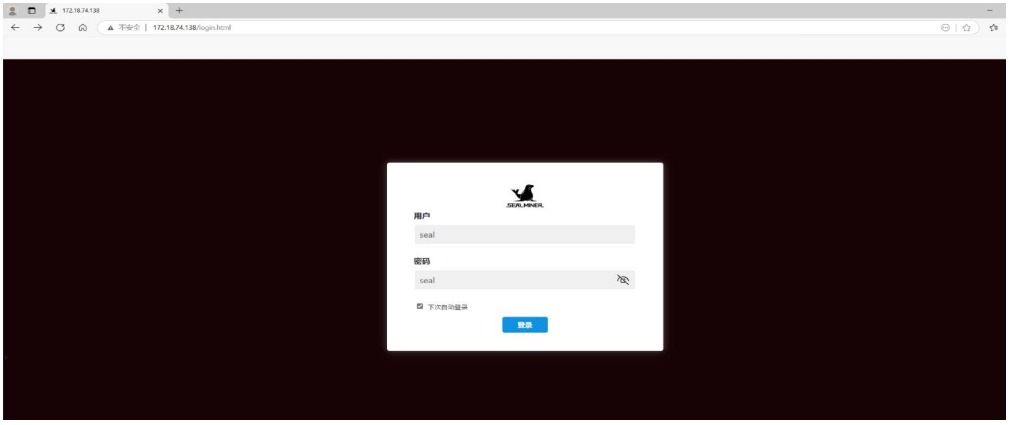
password.)



5.3 Configure the server static IP address (optional)

(1) Log in to the web page

Account: seal, Password: seal



(2) On the Network->Settings page, select "Static" in the Network Type option, change the IP address, mask, gateway, and DNS address to the actual miner address, and click "Save".

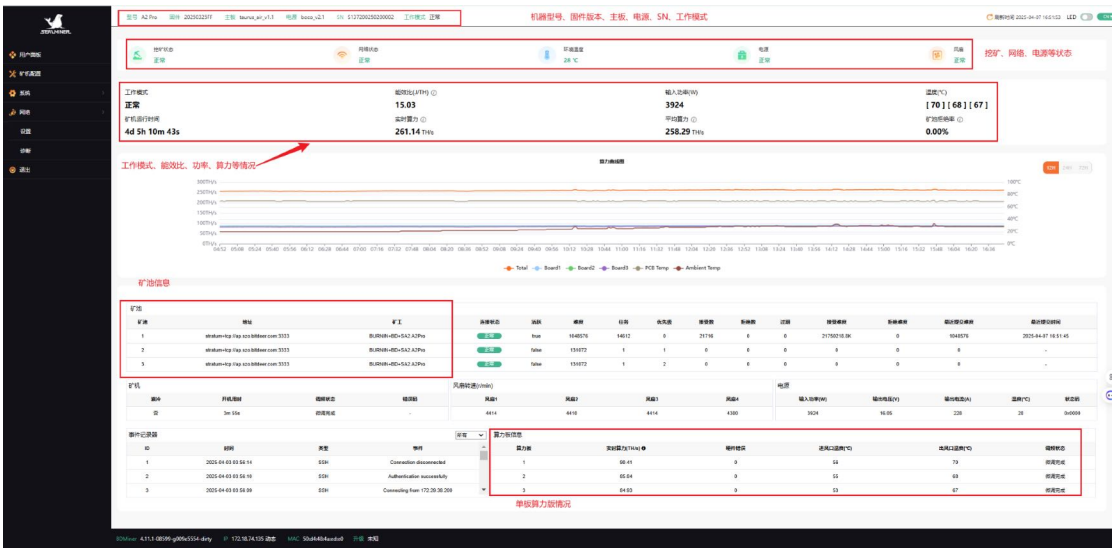


After saving the configuration with a newly set static IP address, you will need to use this new static IP address to log in to the server. Attempting to log in using the previous IP address may result in the page continuously loading until it fails.

6. Server running status check

Once the server is connected to the operational network, log in to the server to check its running status.

1) Within the server interface, select the "User Panel" option to access the "Miner Running Status" interface.
2) Here, you can view the server's overall hash rate, connected mining pool, fan status, individual board hash rates, individual board temperatures, and other operational parameters.(Note: The mining pool address, miner name, and password in the figure below are SEALMINER Demo examples. For actual operations, please refer to your own mining pool address, miner name, and password.)



7. Configuring Server batch data, server status check, upgrading of firmware

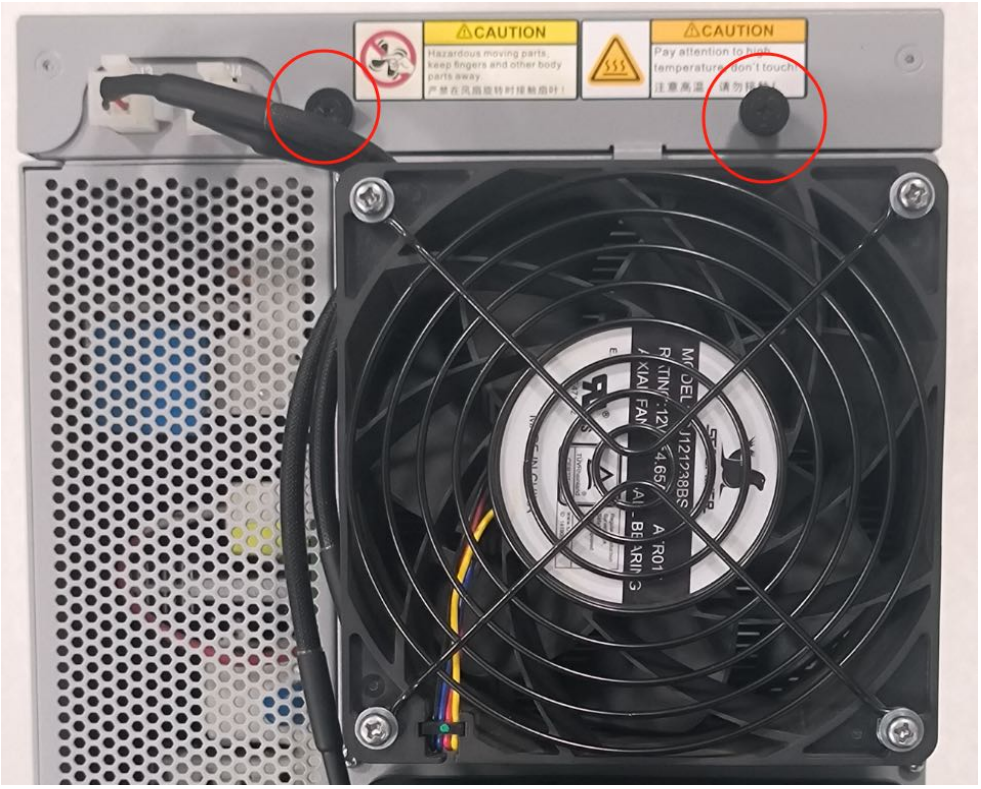
The Sealminer management tool can be used for batch server data configuration, status monitoring, and firmware upgrades. For detailed instructions, please refer to the "Sealminer Management Tool User Manual."

8. Machine Disassembly and Installation

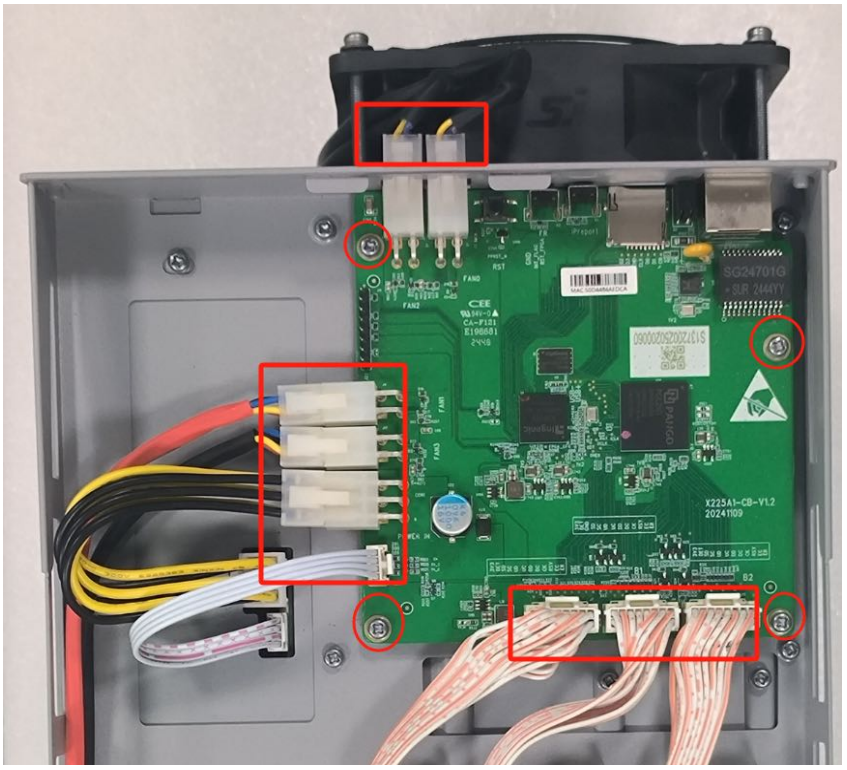
8.1 Disassembly and Installation of the Control Board

8.1.1 Disassembly of the Control Board

(1) First, loosen the two cover screws on the back of the control box, as indicated by the circles in the figure below, and then remove the control box cover;

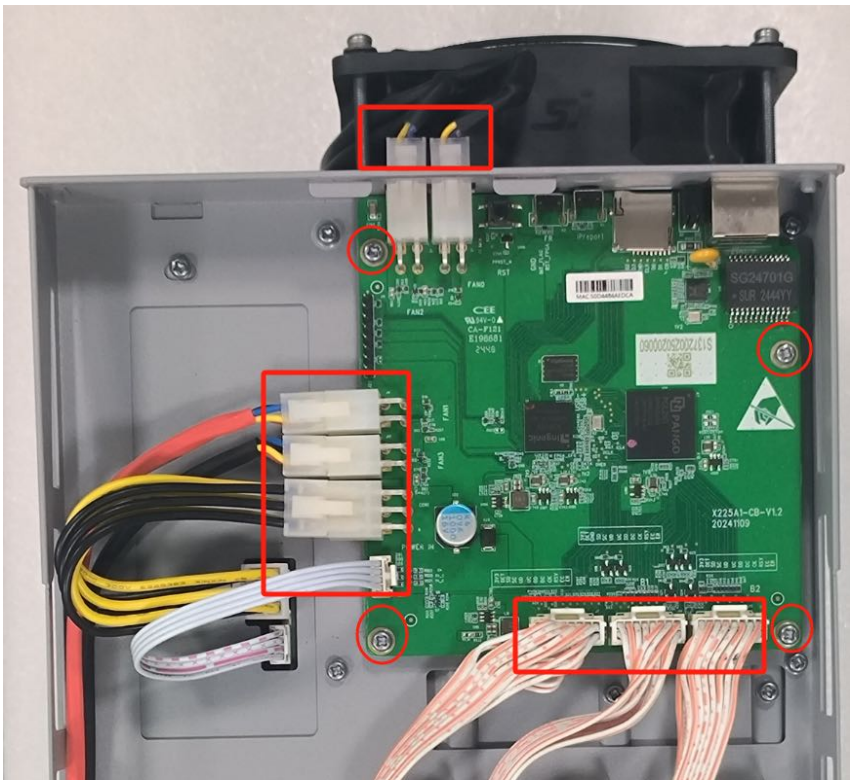


(2) Unplug the fan cable, control board power cable, power IIC control cable, and hash board cable, and remove the corresponding screws at the four corners of the control board as seen in the figure below to remove the control board;



8.1.2 Control board installation

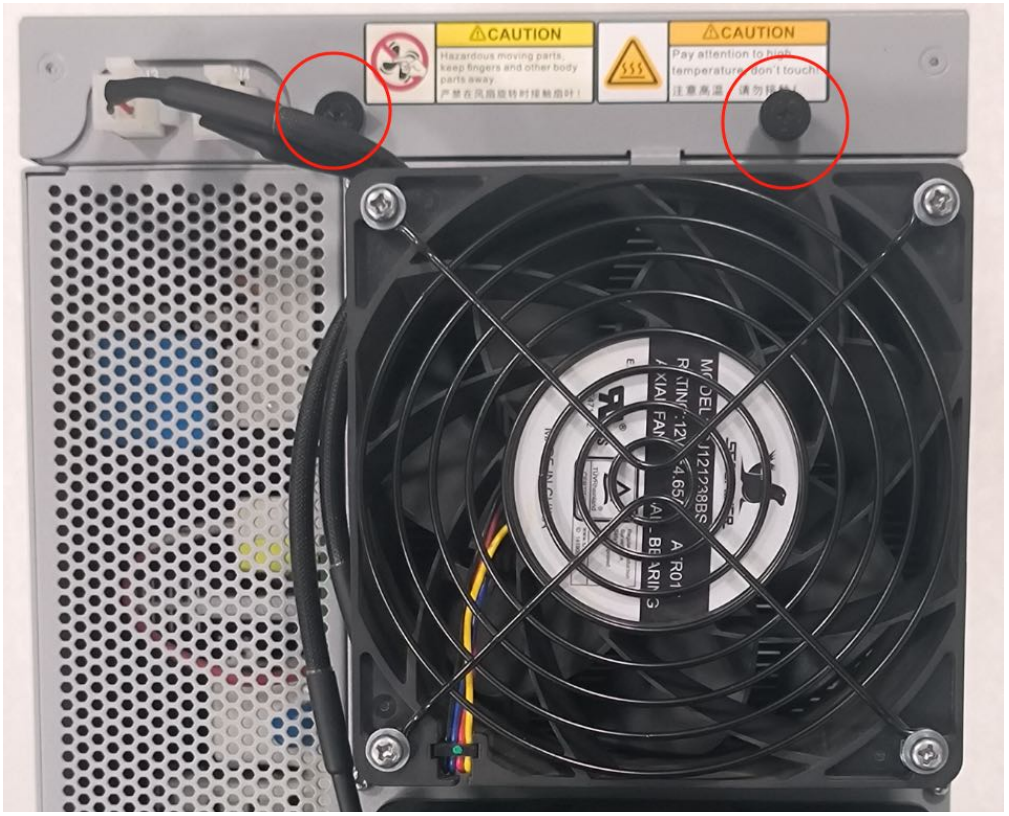
As shown in the figure below, place the control board back into place, tighten the four corner screws, insert the cable firmly, and install the control box cover.



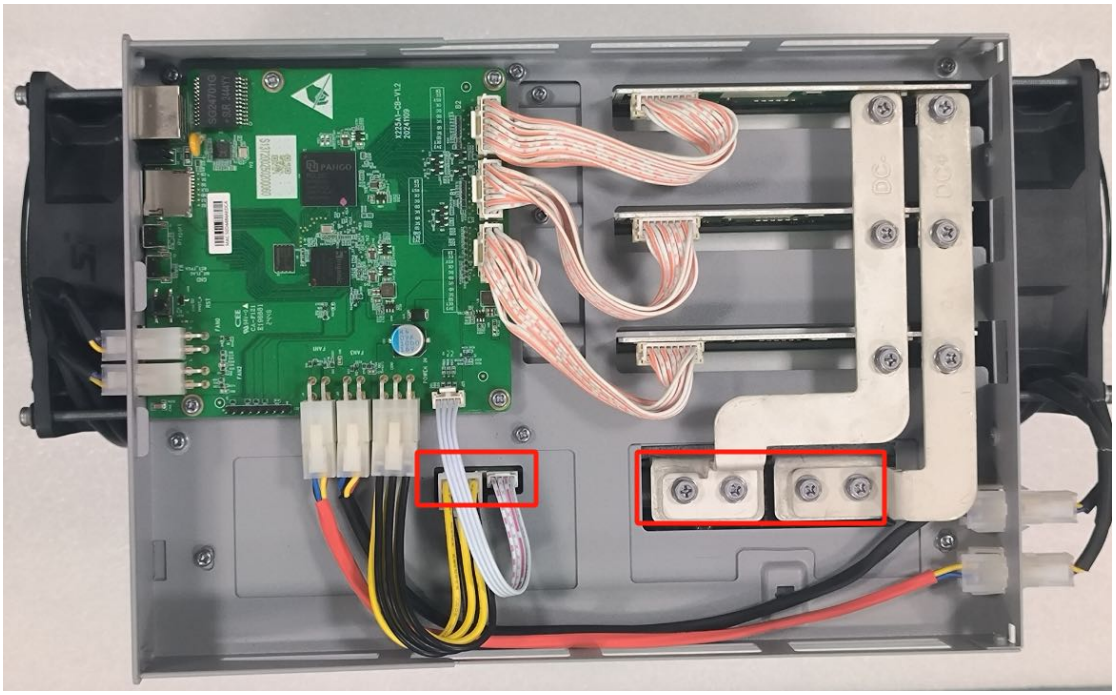
8.2 Disassembly and Installation of the Power Supply

8.2.1 Disassembly of the Power Supply

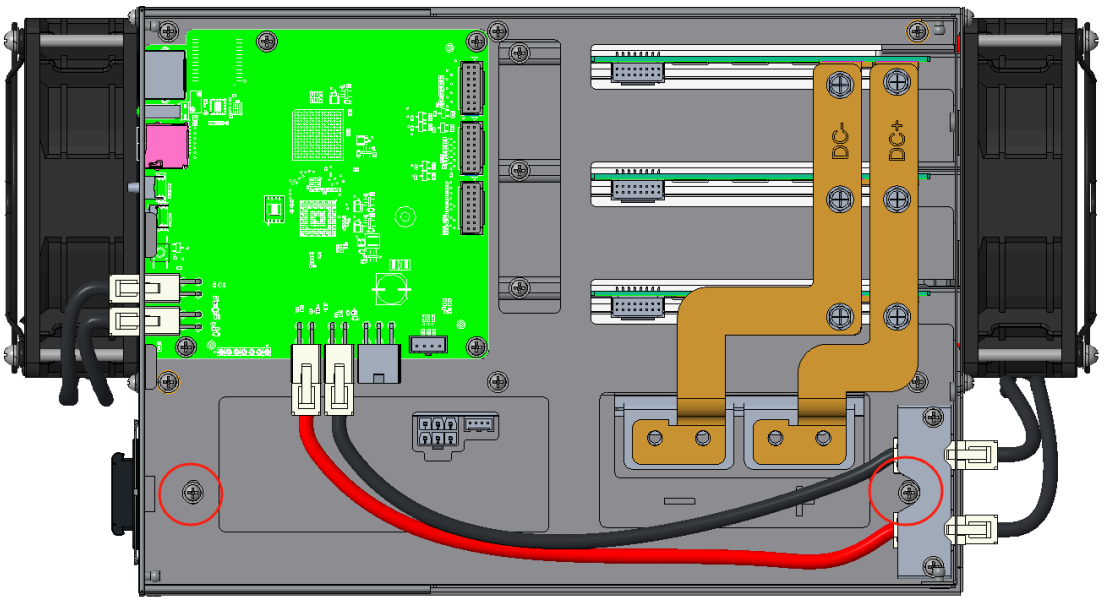
(1) Loosen the two cover screws on the control panel cover and remove the cover;



(2) Unplug the power cord and IIC wire, and remove the 4 screws of the power copper busbar as shown in the figure below;

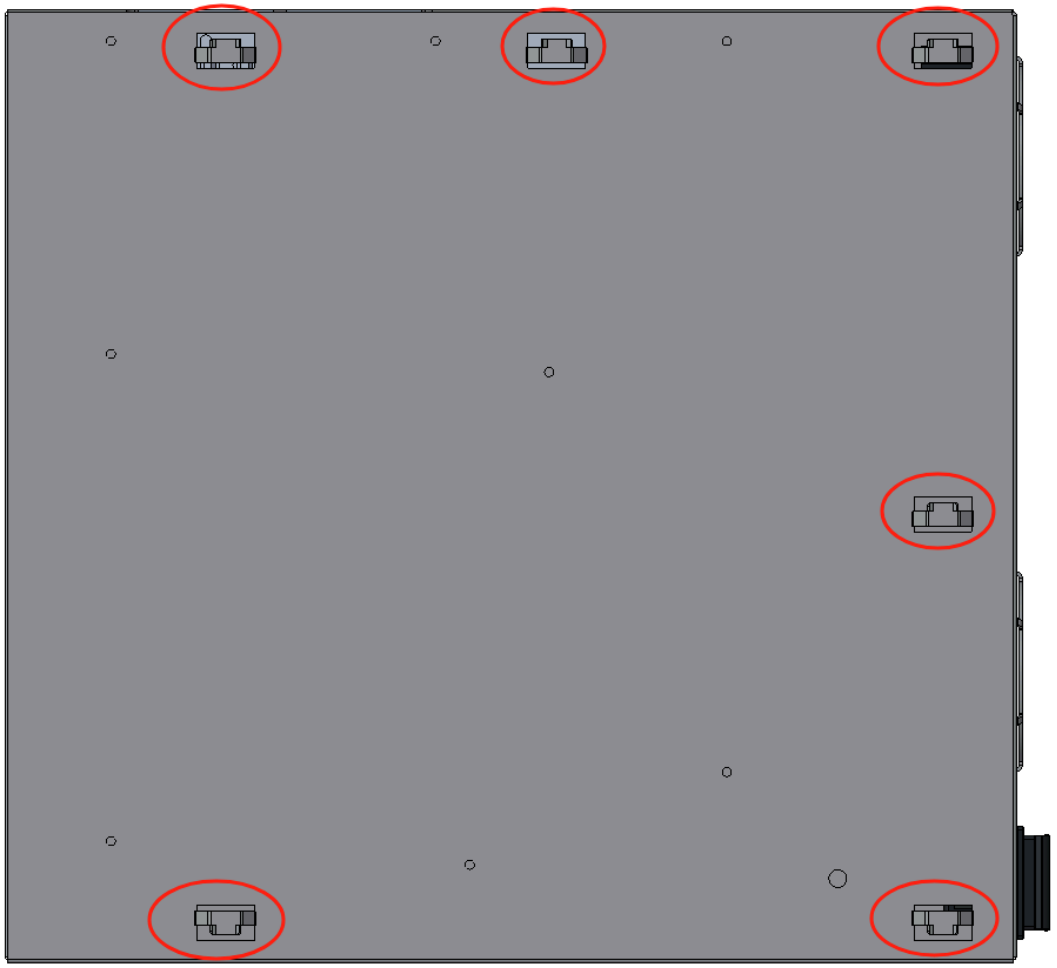


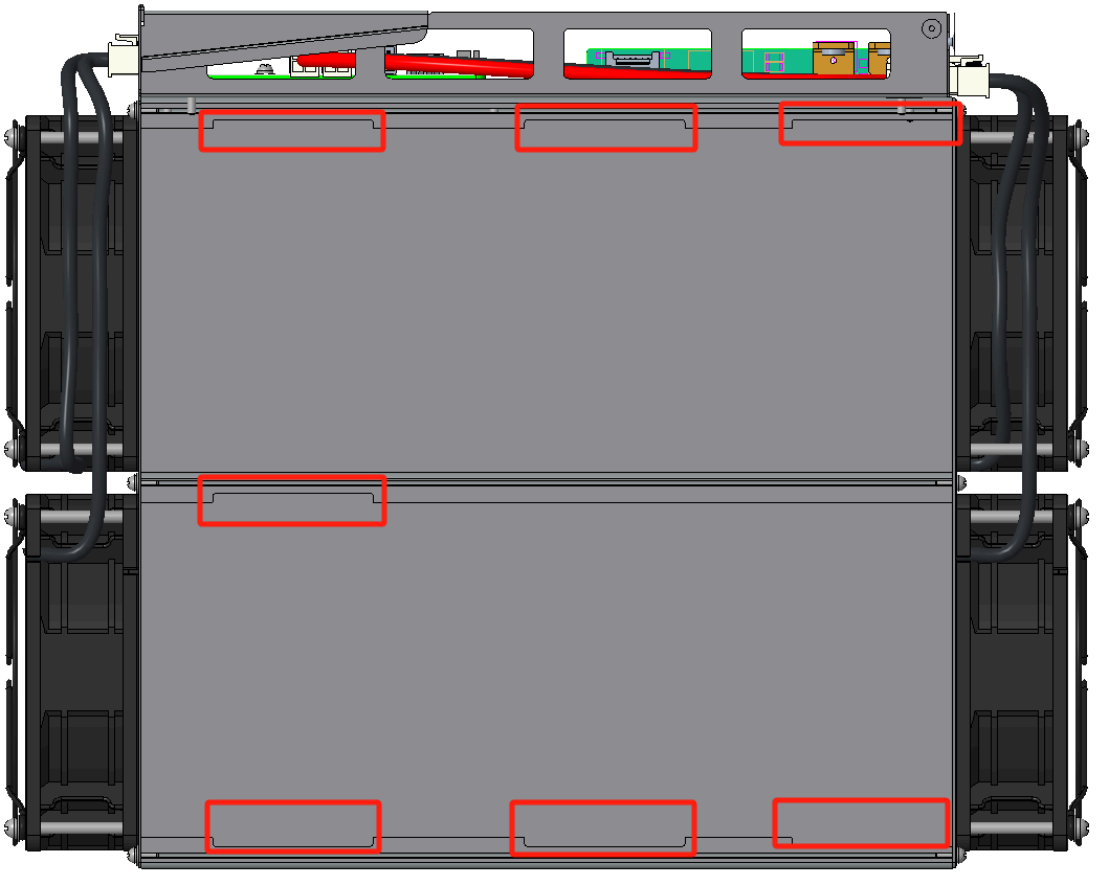
(3) Remove the two fixing screws on the top of the power supply and slide the power supply backward out of the slot;

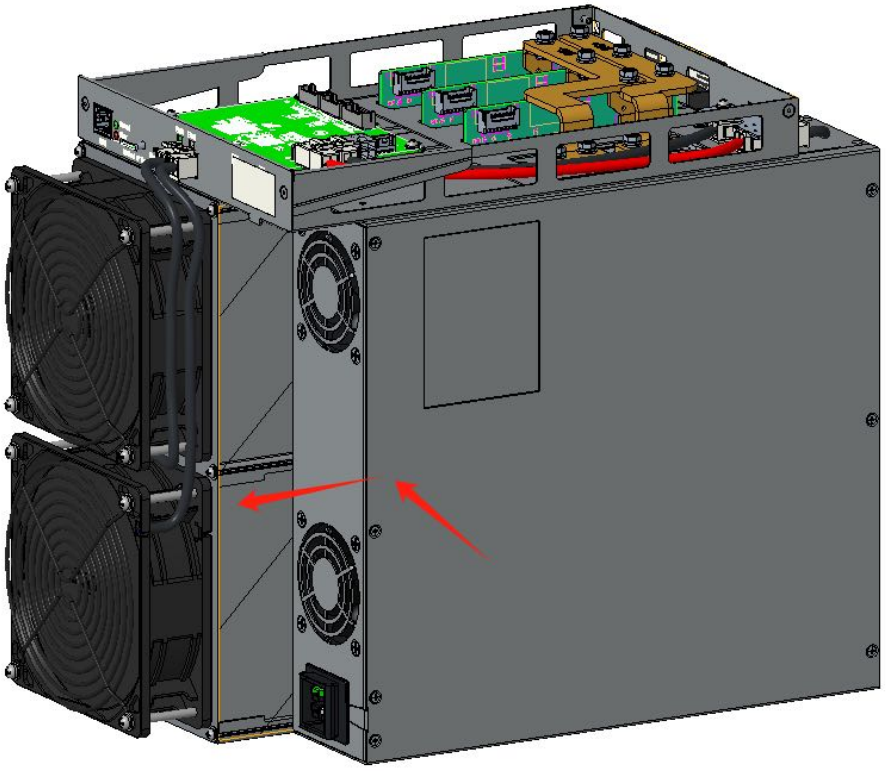


8.2.2 Installation of the Power Supply

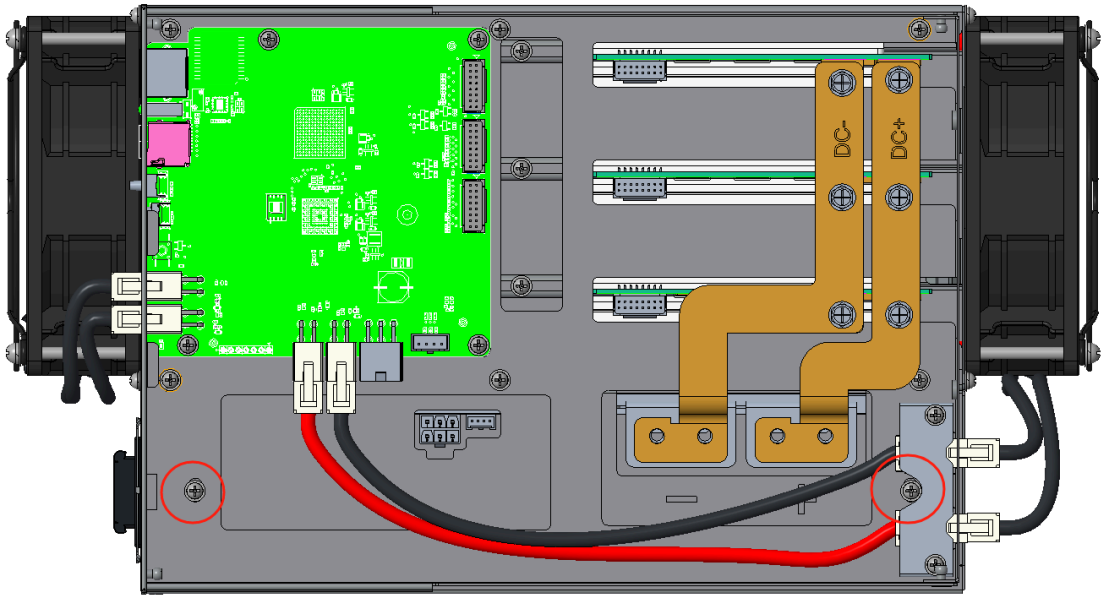
(1) Align the power clip with the chassis slide slot as shown in the figure below and gently push it into place.



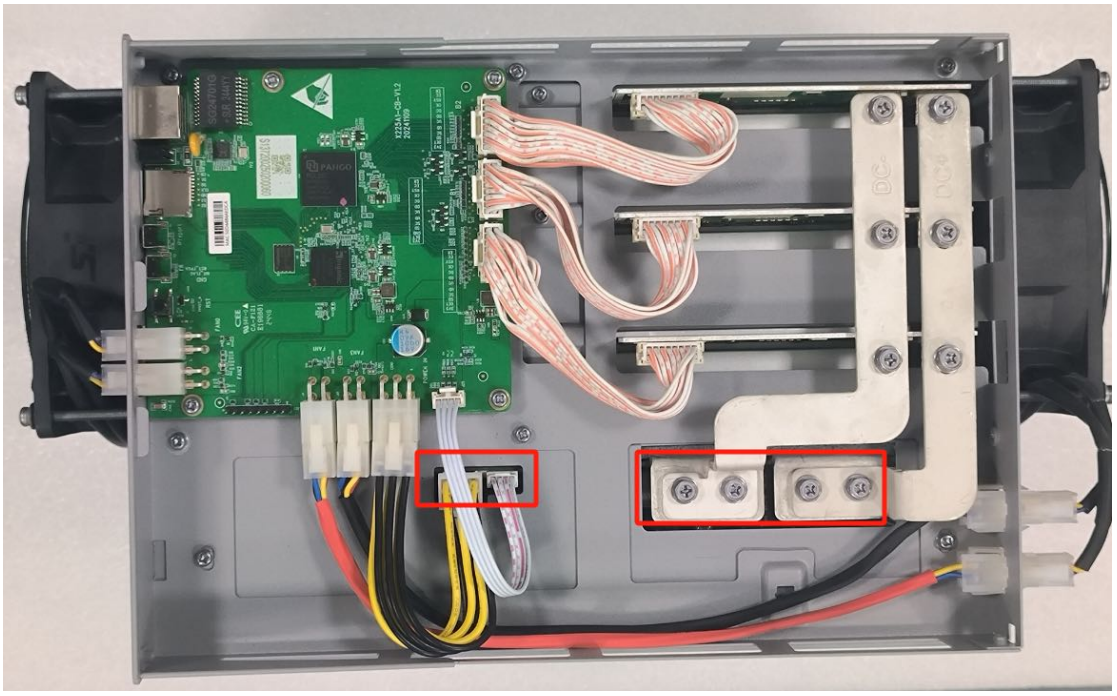




(2) Tighten the two screws shown in the figure below;



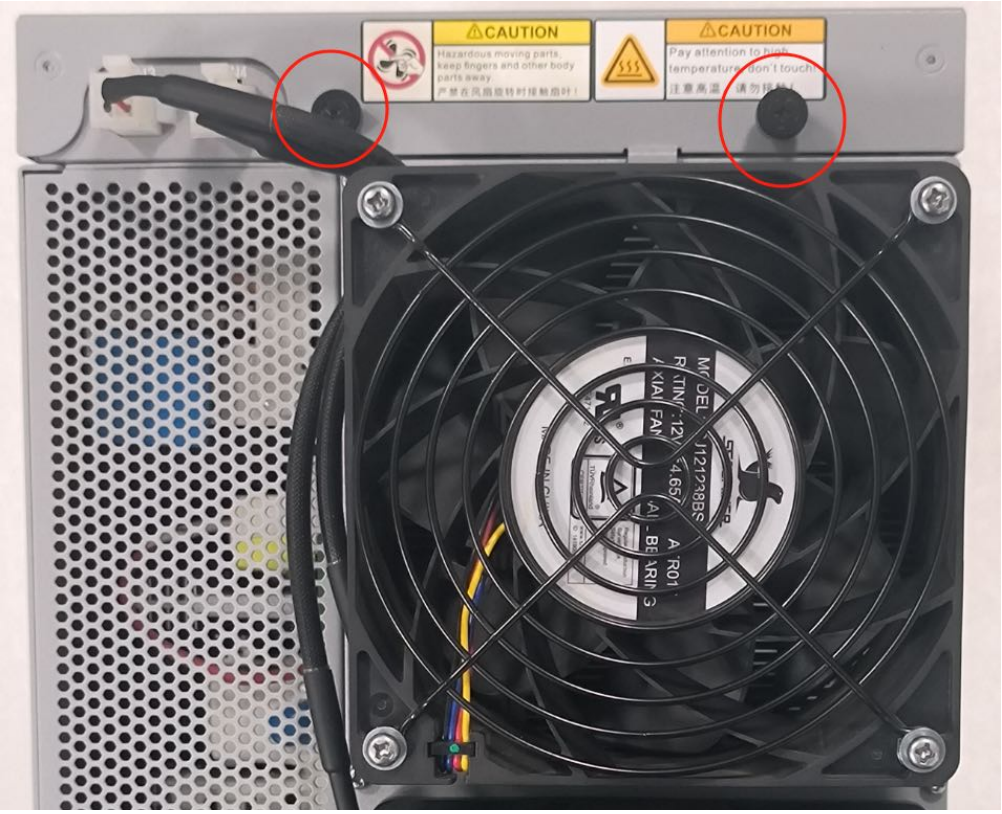
(3) Install the power cable and IIC control cable, then tighten the 4 screws on the copper busbar, and install and lock the control cover.



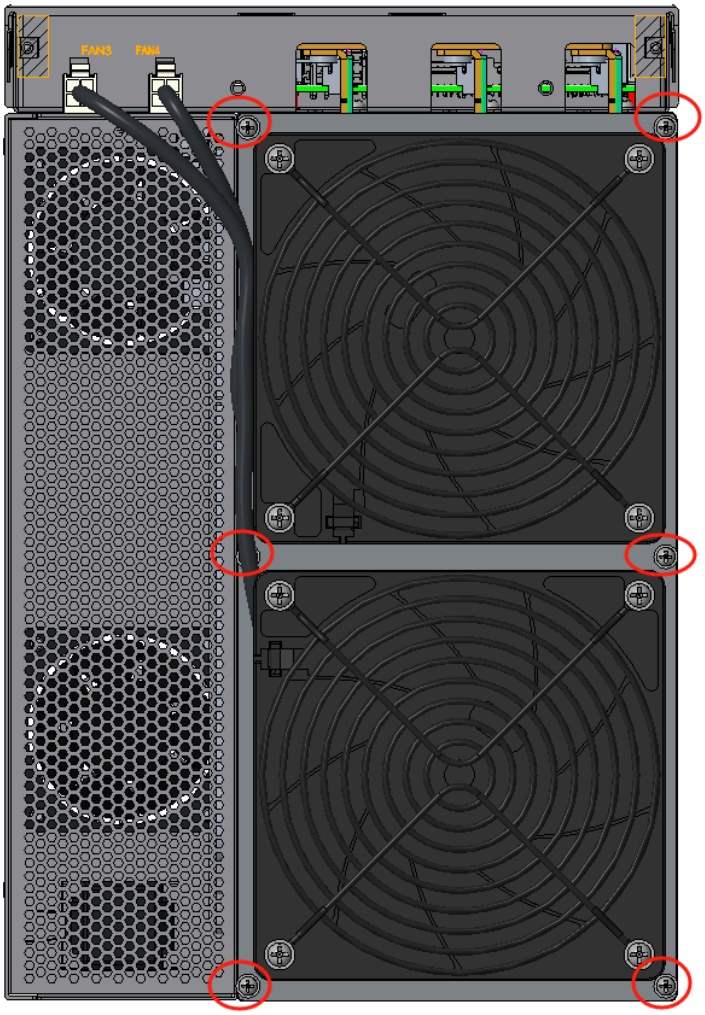
8.3 Disassembly and installation of hashboard

8.3.1 Disassembly of the Hash Board

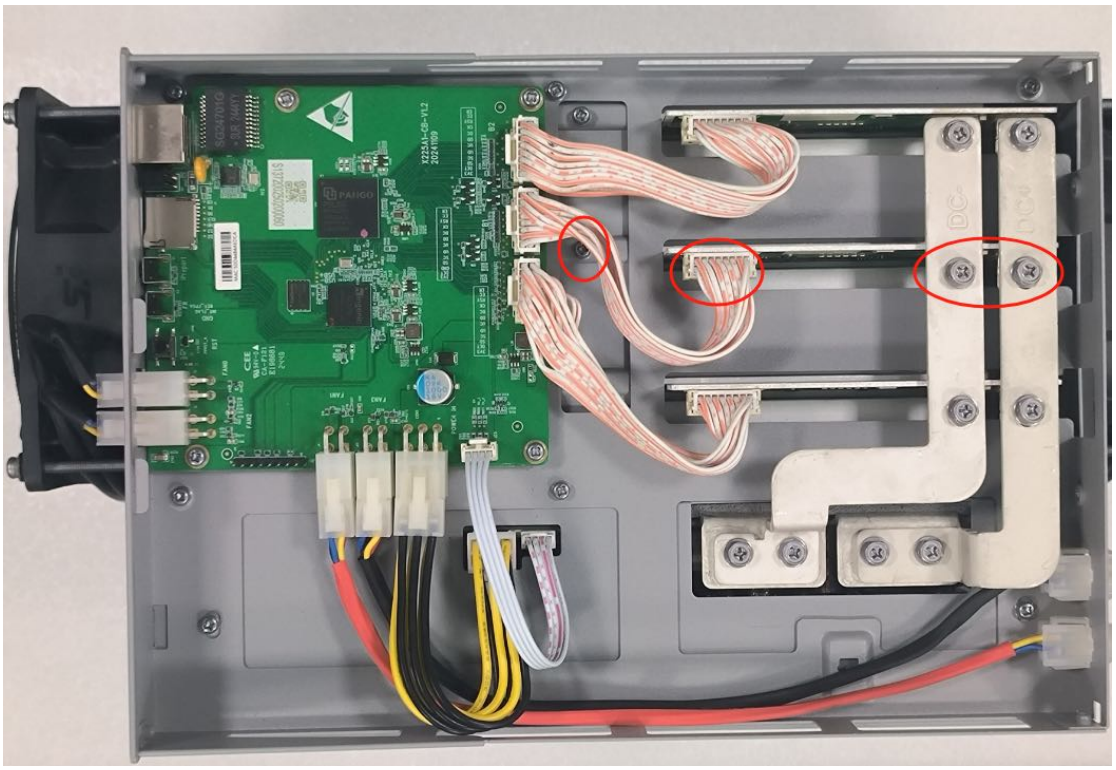
(1) Loosen the two cover screws on the control panel cover and remove the cover;



(2) Remove the six screws on the rear fan fixing plate and remove the rear fan fixing plate as shown in the figure below.

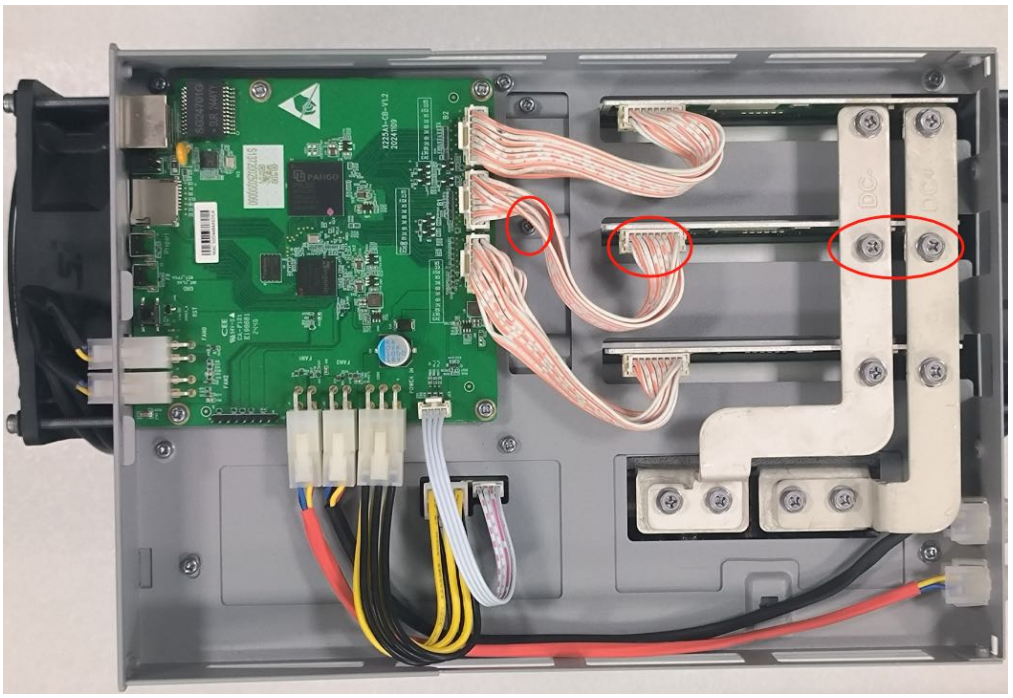


(3) As shown in the figure below, remove the two fixing screws and the two copper busbar screws, unplug the signal cables, gently pull out the hashboard backwards;

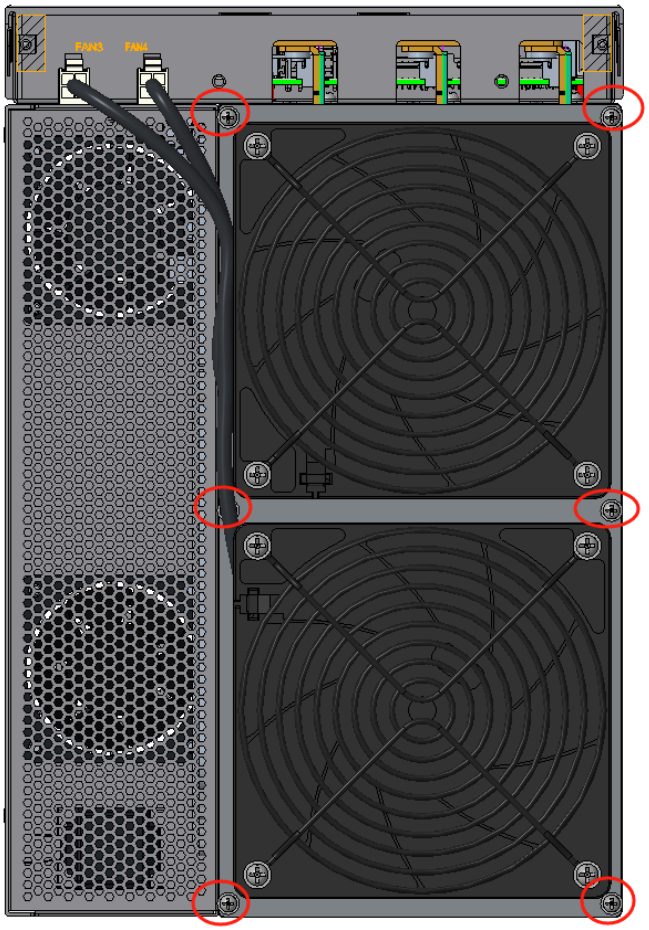


8.3.2 Installation of the Hash Board

(1) As shown in the figure below, insert the hash board through the slide slot, tighten the fixing screws on the hashboard and copper bar screws, and connect the cables ;



(2) Install the rear fan fixing plate and tighten the 6 screws as seen in the figure below;



9. Further Remarks And Considerations

9.1 Equipment Care and Maintenance

After extended operation (3-6 months), it is important to clean the dust from the computing board, control board, wiring, and other components to prevent corrosion from dust buildup. Once the machine is powered off, ensure that the storage environment maintains humidity levels below 60%, and regularly clean dust (every 3-6 months) to prevent corrosion caused by high humidity and dust accumulation.

Note: Try to avoid serious corrosion, dust accumulation, humidity and other conditions that can affect the performance and functionality of the machine.

10. After-Sales Service

At Bitdeer, prioritising user experience is our primary objective. We actively seek customer feedback to enhance our services and are dedicated to assisting customers with their issues. Bitdeer offers a 1-year warranty on A2-Air Cooling Series products, including free repair services within the warranty coverage. By making a purchase, you agree to our warranty terms. However, please note that we will not provide repairs for the following situations or failures:

1. Damage resulting from failure to install, use, maintain, or service the machine according to the requirements outlined in the official documents.
2. Unauthorised disassembly, modification, assembly, or repair without official written or electronic authorization, leading to damage to the product.
3. Damage or loss caused by mishandling, misuse, abuse, improper operation, incorrect installation, or maintenance and storage that do not comply with regulations.
4. Product damage arising from operating the mining machine in environments that do not meet the specified requirements, including but not limited to high humidity, corrosive environments, electrical surges, extreme temperatures, abnormal voltage or current (including surges, impacts, and instability), excessively low or high AC voltage.
5. Damage to the entire machine, the panel, or components due to crushing, breaking, burning, or falling, resulting from improper operation or similar issues.
6. Product damage caused by overvoltage, undervoltage, or leakage.
7. Damage or loss resulting from unforeseen natural disasters, including but not limited to floods, fires, earthquakes, tsunamis, and lightning strikes.

8. Disassembly or modifications made by anyone other than us or our authorised service agencies.
9. Product failure or damage resulting from the use of accessories, parts, or components, such as power supplies, that are not manufactured by us or our authorised agencies and do not meet our specified parameters.
10. Failure or damage resulting from the use of unauthorised firmware or drivers, including but not limited to unauthorised overclocking firmware.
11. Altered, defaced, or removed SN labels.
12. Use of non-original or mixed boards: if some or all of the computing boards, control boards, or power supplies are not original to the machine or if there is any situation preventing us from verifying the authenticity of these components.
13. Situations not caused by our company that make it impossible for us to determine whether the product is still under warranty.

11. After-sales warranty Fee terms

1. If your product is deemed non-repairable, or if you opt not to repair or return a scrapped or mixed-board product, you will be responsible for the round-trip shipping costs associated with the repair order. Additionally, for products requiring repair, you will be responsible for the shipping costs to send the product to our designated address, while we will cover the shipping costs to send the repaired or replaced product back to your designated address. We do not cover any other costs, such as tariffs.
2. Please ensure that the product is sent to our designated address via a prepaid postage service. We cannot accept items sent with unpaid postage or to non-designated addresses (including logistics pick-up points), and any consequences of these errors will be borne by you.
3. The repaired or replaced product will be shipped according to the delivery information you or your designated contact provided in the work order. If the provided delivery information is incorrect or incomplete, you will be responsible for any additional costs incurred.

4. In the event of a Dead on Arrival (DOA) situation or secondary repair where the product is non-repairable or not eligible for free repair, you may apply for a freight subsidy after sending the product to us by post. The subsidy amount will not exceed the official logistics fee standard (insurance costs are the customer's responsibility). To qualify, you must provide us with a valid receipt for the shipping costs. The freight receipt must not be forged, altered, smeared, or tampered with in any way, and it must clearly show the total freight amount, which must align with the official logistics fee standard. If the receipt does not meet these requirements, we reserve the right to deny part or all of your freight subsidy request.

5. Please ensure that products eligible for a freight subsidy are mailed separately. If you send a package that includes both products eligible for a subsidy and those that are not, we will not be able to calculate the freight for the eligible items individually, and therefore, we will not issue any subsidy for that package.

6. The risk of damage or loss during the return process transfers to you once the product is handed over to the logistics company. If any damage or loss occurs during transit, you will be responsible for resolving any disputes directly with the logistics provider.

7. Any matters not covered in this clause shall be handled in accordance with the relevant regulations of Bitdeer. Bitdeer reserves the final right of interpretation of this clause.

12. Regulatory and Compliance information

12.1 United States Federal Communications Commission (FCC)

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

Class A Digital Device Notice

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at their own expense.

Mandatory Shielding & Grounding

To ensure continued compliance with FCC Class A emission limits, shielded cables must be used with this equipment. Use of non-shielded

cables may result in interference exceeding legal limits. The equipment must be connected to a power outlet with a protective earthing (grounding) connection.

FCC Caution

Any changes or modifications not expressly approved by Bitdeer could void the user's authority to operate the equipment.

Supplier's Declaration of Conformity (SDoC)

FCC Supplier's Declaration of Conformity

Trade Name: SEALMINER

Model Number: A2 Pro

Responsible Part---U.S. Contact Information

Company: Bitdeer Sales (USA) Inc.

Street Address: 2880 Zanker Rd., Suite 205

City, State: San Jose, CA

Zip Code: CA 95134

Telephone number: +1 408-451-5200

Internet contact information: https://www.bitdeer.com/

FCC Compliance Statement:

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

The FCC Supplier's Declaration of Conformity (SDoC) provided herein applies to the following models within the Air Series: A2, A2 Pro, A3 Air, A3 Pro Air, A4 Pro Air.

12.2 Innovation, Science and Economic Development (ISED) Canada

(English) This Class A digital apparatus complies with Canadian ICES-003 Issue 7.

(Français) Cet appareil numérique de la classe A est conforme à la norme NMB-003, version 7, du Canada.

Compliance Statement (English):

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Déclaration de Conformité (Français):

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

1. L'appareil ne doit pas produire de brouillage;
2. L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Labelling Identification:

CAN ICES-003 (A) / NMB-003 (A)

12.3 European Union (CE) & WEEE

SEALMINER

CE

EU Declaration of Conformity

We
Manufacturer's Name: Bitdeer Semiconductor Technology Pte. Ltd.
Manufacturer's Address: 63 Chulia Street, #15-01 OCBC Centre East Singapore 049514, Singapore

EU AR Name: Bitdeer Norway AS, c/o Javelin Corporate Services AS
EU AR Address: Vogts gate 64, 0477 Oslo, Norway
Contact: aftersales@bitdeer.com | www.bitdeer.com

declare under sole responsibility for issuing this declaration of conformity in relation to the following product:

Trademark: SEALMINER
Type of Product: Server
Model: A2 Pro
Batch Number:
Serial Number:
Product Description:
High-efficiency SHA-256 ASIC Miner for BTC/BCH/BSV coin mining.
Rating: Input: 220-277VAC, 50/60Hz, 27A MAX



For the above given Product is hereby declared that it conforms to the essential requirements set out in community harmonization legislation mentioned below:

- Electromagnetic Compatibility(EMC) Directive (2014/30/EU)
- Low Voltage Directive (LVD) (2014/35/EU)
- Restriction of the use of certain hazardous substances (RoHS) directive (2011/65/EU) and its amendment Directive (EU) 2015/863

Reference to the relevant harmonised standards used, or references to the specifications in relation to which conformity is declared:

EMC standard:	EN 55032:2015+A11+A1 EN IEC 61000-3-11:2019 EN 61000-3-12:2011+A1 EN 55035:2017+A11
Safety standard:	EN 62368-1: 2014 + A11:2017
Chemical standard:	EN IEC 63000:2018

Signed for and on behalf of:

Singapore	23 Mar 2026	
Place of Issue	Date of Issue	Name, Function, Signature

The EU Declaration of Conformity provided in this section is representative of the A2 Pro model. For the Declaration of Conformity for other models within the Air Series, please contact Bitdeer Customer Support.

WEEE (Waste Electrical and Electronic Equipment)

The WEEE symbol below indicates that this product must not be disposed of with household waste. In accordance with EU Directive 2012/19/EU, this product must be handed over to an authorized collection point for the recycling of electrical and electronic equipment (EEE).

Symbol	Instructions
	The symbol shown (crossed-out wheelie bin) is on the product or its packaging, which indicates that this product must not be disposed of with other waste. Instead, it is the user's responsibility to dispose of their waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. The separate collection and recycling of your waste equipment at the time of disposal will help to conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For more information about where you can drop off your waste equipment for recycling, contact your local waste disposal service or Bitdeer representative for recycling information.

12.4 Safety & Power Specifications

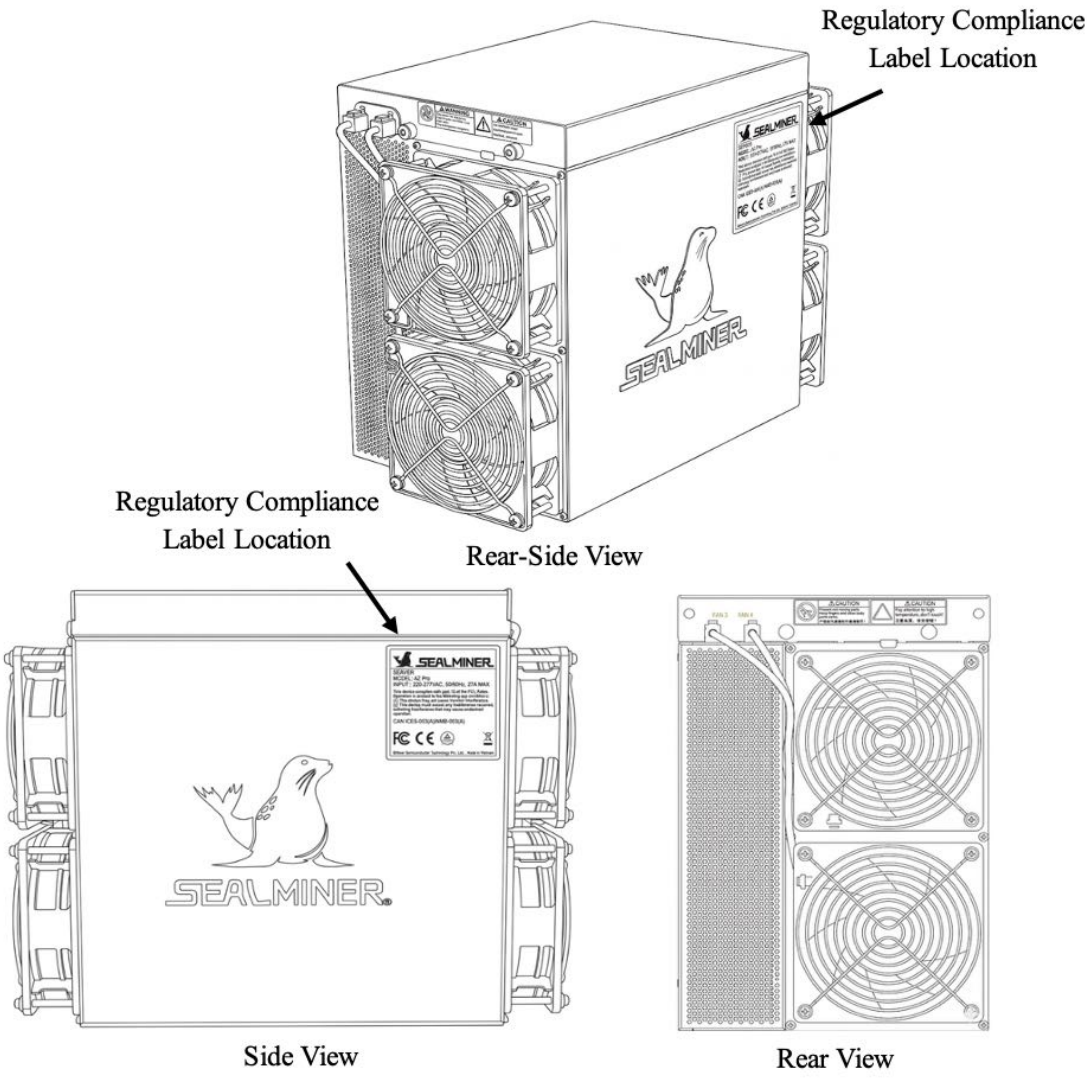
- **Installation:** The socket-outlet must be installed near the equipment and remain easily accessible. The upstream circuit breaker in the PDU serves as the disconnection device, and its rated breaking current must be 20A.
- **Power Cord:** Use only the power cords provided or specified by Bitdeer. Connect the server to the PDU using the provided

1.2-meter power cord, ensuring the connector is fully seated in the server's power port (labeled as component 8 in the Component Reference). To ensure continued compliance with emission limits, shielded cables must be used.

- **Earthing:** The equipment must be connected to a power outlet with a protective earthing (grounding) connection to prevent electric shock.
- **Operating Environment:** Industrial use only. Not intended for use in locations where children are likely to be present.

12.5 Product Identification & Labelling

The regulatory label is affixed to the chassis of the device (typically on the side panel). This label contains critical information including the Model Number, Serial Number, and required certification marks (FCC, CE, IC, cTUVus & WEEE).



Sample Regulatory Label: Below is a representation of the compliance label found on the device:

Trademark: **Type of Product:** **Model:** **Input:** [Voltage/Current Specs]
FCC Compliance Statement: This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference. (2) This device must accept any interference received, including interference that may cause undesired operation.

CAN ICES-003 (A) / NMB-003 (A)
[FCC Logo] [CE Logo] [cTUVus Logo] [WEEE Bin]
Manufacturer Name **Manufacturer Address** **EU Authorized Representative Name** **EU Authorized Representative Address** **Contact information** **Made in [Country]**